

Mathematics, 2018 EC, entrance exam, version 3

Grade 12 · Mathematics · 2018 E.C.

Time allowed: 120 minutes

Questions: 50

Total marks: 50

Name _____ Grade _____ ID No. _____ Date _____

1. Which of the following statement defines an arithmetic progression? (1)
 - A. It is a sequence in which the ratio between consecutive terms is a positive constant.
 - B. It is a sequence in which the difference between consecutive terms is a non-zero constant.
 - C. It is a sequence in which the difference between consecutive terms is a constant.
 - D. It is a sequence in which the ratio between consecutive terms is a non-zero constant.
2. The simplified form of the expression $(\sqrt{32} + \sqrt{48}) / (\sqrt{8} + \sqrt{12})$ is equal to: (1)
 - A. $\sqrt{2}$
 - B. 4
 - C. 1
 - D. 2
3. Which one of the following pairs of numbers are irrational numbers between 11 and 13? (1)
 - A. $\sqrt{120}$ and $\sqrt{168}$
 - B. $3+\sqrt{81}$ and $2\sqrt{37}$
 - C. $2+\sqrt{112}$ and $3\sqrt{17}$
 - D. $\sqrt{125}$ and $\sqrt{175}$
4. If a polynomial function $f(x)=x^3+ax^2+bx+3$ is divided by $x+2$ the remainder is 15 and $x-1$ is a factor of $f(x)$, then what are the values of a and b respectively? (1)
 - A. -6,2
 - B. 6,2
 - C. 2,6
 - D. 2,-6
5. Which of the following ordered pairs gives the coordinates of the point that divides a line segment with endpoints A(-1,5) and B(5,2) in the ratio 1:2? (1)
 - A. (3,3)
 - B. (-3,-3)
 - C. (-4,-1)
 - D. (1,4)
6. Given sets $A = \{x: x^2-12x+20=0\}$ and $B = \{x: x^2+5x-14=0\}$, then relative complement of A with respect to B ($B \setminus A$) is: (1)
 - A. $\{-7\}$
 - B. $\{10\}$
 - C. $\{-7,10\}$
 - D. $\{-10,7\}$
7. Which one of the following statements is necessarily true? (1)
 - A. If a feasible region of a linear programming problem is unbounded, then there is no optimal value.
 - B. If a feasible region of a linear programming problem is bounded, then the objective function has either maximum value or minimum value.
 - C. If a feasible region of a linear programming problem is bounded, then the objective function must have optimal value(s).
 - D. At corner points of a feasible region of any linear programming problem there are always optimal values.

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8. A group of 30 men participate in a survey on language skills. The number of men who know both English and Afan Oromo equals the number who know neither. The number who know English is 4 more than those who know Afan Oromo. How many know Afan Oromo? (1)
- A. 26
B. 14
C. 13
D. 11
-
9. Tola purchased a TV stand for Birr 5000 and paid Birr 250 for transportation. If he sold it for Birr 5075, then which statement is true? (1)
- A. Tola gain 3.33%
B. Tola gain Birr 175.
C. Tola gain Birr 75.
D. Tola loss 3.33%
-
10. The cost of an electronic device reduces at the rate of 5% per annum. If its present cost is Birr 32490, then what was its cost before two years? (1)
- A. 36,000
B. 37,200
C. 35,400
D. 29,469
-
11. If three positive numbers are in the ratio 1:3:5 and the sum of their squares is 875, then which is the sum of the three numbers? (1)
- A. 45
B. 75
C. 90
D. 99
-
12. If a man saves Birr 135 in the first year, Birr 150 in the second year, Birr 165 in the third year and so on, then in what time (in years) is his total saving Birr 5,550? (1)
- A. 362
B. 37
C. 20
D. 360
-
13. If the 6th and 11th terms of an arithmetic progression are 52 and 112 respectively, then the common difference is: (1)
- A. 12
B. 15
C. 20
D. 4
-
14. If the first term of an arithmetic sequence is 3 and the sum of its first four terms is equal to one-fifth of the sum of the next four terms, then the sum of the first 20 terms is: (1)
- A. -1040
B. -1080
C. -1200
D. -2160
-
15. What is the absolute maximum of $f(x)=2x^3-6x^2-48x+7$ on $[-6,8]$? (1)
- A. 353
B. 263
C. 63
D. -353
-
16. If $\int_0^k (2kx - x^2) dx = 18$, then which is the value of k ? (1)
- A. -3
B. 0
C. 9
D. 3
-
17. If F and f are defined on $[a,b]$ such that $F'(x)=f(x)$ for all x , then $\int_a^b f(x)dx$ is equal to: (1)
- A. $F'(a)-F'(b)$
B. $F'(b)-F'(a)$
C. $F(a)-F(b)$
D. $F(b)-F(a)$

18. Which is the area enclosed by $f(x)=4x^3$ and x-axis between $x=-2$ and $x=3$? (1)
- A. 65
B. 81
C. 97
D. 107
-
19. On interval $[0,2\pi]$, which θ satisfy $2\cos\theta + \sqrt{3} = 0$? (1)
- A. $4\pi/3, 5\pi/3$
B. $2\pi/3, 4\pi/3$
C. $5\pi/6, 7\pi/6$
D. $8\pi/6, 11\pi/6$
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20. The solution set of $\log(x+4) + \log 4 = \log 48 - \log x$ is: (1)
- A. $\{2\}$
B. $\{-6, 2\}$
C. $\{-6\}$
D. $\{-2, 6\}$
-
21. If $f(x)=\sin x$ and $g(x)=\cos x$, which is NOT true? (1)
- A. Both functions have the same range.
B. The domain of both functions is $\mathbb{R}$.
C. The value of both functions can be less than -1.
D. They have the same period.
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22. Let $f(x) = (2x-3)/(x+1)$, $x \neq -1$. Which is the inverse? (1)
- A. $f^{-1}(x) = (x+3)/(2-x)$, $x \neq 2$
B. $f^{-1}(x) = (x+1)/(2x-3)$, $x \neq 3/2$
C. $f^{-1}(x) = (3-x)/(2x+1)$, $x \neq -1/2$
D. $f^{-1}(x) = (2x+3)/(x-1)$, $x \neq 1$
-
23. The domain and range respectively for the power function $f(x) = (1/3)x^{-5}$ are: (1)
- A. $\mathbb{R} \setminus \{0\}$ and $(0, \infty)$
B. $\mathbb{R} \setminus \{0\}$ and $\mathbb{R} \setminus \{0\}$
C. $(0, \infty)$ and $(0, \infty)$
D. $[0, \infty)$ and $[0, \infty)$
-
24. Which is the largest interval on which $f(x) = -2x^3 - 9x^2 - 12x + 1$ is increasing? (1)
- A. $(-\infty, -2] \cup [-1, \infty)$
B. $[-2, -1]$
C. $(-\infty, -2] \cup [1, \infty)$
D. $[-2, \infty)$
-
25. Suppose mean=32, median=30.5, mode=29.5. Which is true? (1)
- A. Distribution is positively skewed.
B. Bowley's coefficient of skewness $\beta < 0$.
C. Distribution is negatively skewed.
D. Pearson's coefficient of skewness is zero.
-
26. Which one is non-random sampling? (1)
- A. Stratified Sampling
B. Convenience Sampling
C. Systematic Sampling
D. Cluster Sampling
-
27. A company manufactures sweaters type A and B. Cost: A=360, B=120. Can make at most 300 sweaters and spend at most 72,000 a day. Number of type B cannot exceed type A by more than 100. Profit: A=200, B=120. Which linear programming model maximizes profit? (1)
- A. Maximize $P=200x+120y$ subject to: $x+y \leq 300$, $360x+120y \leq 72000$, $y \leq x+100$, $x, y \geq 0$
B. Maximize $P=120x+200y$ subject to: ...
C. Maximize $P=200x+120y$ subject to: $x+y \geq 300$, ...
D. Maximize $P=200x+120y$ subject to: $x+y \leq 300$, $360x+120y \leq 72000$, $y \leq x-100$, $x, y \geq 0$

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28. A circular metal plate expands. Radius increases at 0.5 cm/sec. Rate of increase of area when radius=20 cm? (1)
- A. $80\pi \text{ cm}^2/\text{sec}$ B. $20\pi \text{ cm}^2/\text{sec}$
C. $10\pi \text{ cm}^2/\text{sec}$ D. $200\pi \text{ cm}^2/\text{sec}$
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29. On selling an article for Birr 450, a dealer lost 10%. The cost price is: (1)
- A. Birr 405 B. Birr 500
C. Birr 550 D. Birr 454.55
-
30. If in increasing geometric progression, the 6th term is 32 and 8th term is 128, then common ratio is: (1)
- A. -2 B. 4
C. 2 D. ± 1
-
31. What is the image of point (2,-3) when reflected about the line $y = -x$? (1)
- A. (-3,2) B. (2,3)
C. (3,-2) D. (-2,-3)
-
32. If $Q_3=50$ and $Q_1=35$, then coefficient of quartile deviation is: (1)
- A. $3/17$ B. $15/2$
C. $5/17$ D. $17/3$
-
33. Consider matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 6 & 1 & 9 \\ 4 & 0 & 5 \end{bmatrix}$. Which is $\det(A)$ and the cofactor of entry a_{32} ? (1)
- A. 5, 0 B. -19, -9
C. -19, 7 D. 5, 9
-
34. If $u = i - j$ and $v = 2i + j$, then cosine of angle between u and v is: (1)
- A. $3\sqrt{10}/10$ B. $\sqrt{10}$
C. $3\sqrt{10}/10$ D. $\sqrt{10}/10$
-
35. Bona's income Birr 180,000 per year. Pays 20% house rent, then 15% of remainder on child's education. Money left? (1)
- A. Birr 144,000 B. Birr 158,000
C. Birr 21,000 D. Birr 122,400
-
36. Areas of two similar triangles are 144 cm^2 and 81 cm^2 . If one side of first triangle is 6 cm, then corresponding side of second triangle is: (1)
- A. $81/4 \text{ cm}$ B. $9/2 \text{ cm}$
C. $3/2 \text{ cm}$ D. $9/4 \text{ cm}$
-
37. The table shows marks of 100 students in Mathematics and Physics. Mean: Math=20, Physics=25; SD: Math=2, Physics=3. Which subject shows highest variability? (1)
- A. Mathematics B. Physics
C. Both equal D. Cannot determine
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38. A frustum of a right circular cone with base radii 3 cm and 4 cm, slant height 6 cm. Total surface area? (1)
- A. $67\pi \text{ cm}^2$ B. $74\pi \text{ cm}^2$
C. $56\pi \text{ cm}^2$ D. $36\pi \text{ cm}^2$

39. Solution set of $\sqrt[3]{64^x(x+1)} = (1/16)^{(2x-3)}$ is: (1)
- A. $\{-1\}$ B. $\{-2\}$
C. $\{2\}$ D. $\{1\}$
-
40. Which is true about the graph of $f(x) = (x-2)/(x^2-5x+6)$? (1)
- A. $x=2$ and $x=3$ are vertical asymptotes B. The x -axis is its horizontal asymptote
C. $y=0$ is its oblique asymptote D. The x -intercept is $(2,0)$
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41. Let A be 3×3 matrix and k scalar. Which is true? (1)
- A. If B obtained by multiplying one row by k , then $|B|=|A|$
B. If B obtained by interchanging two rows, then $|B|=|A|$
C. If $B = kA$, then $|B| = k^3|A|$
D. If B obtained by adding k times one row to another, then $|B|=k|A|$
-
42. Let $u = i+3j+2k$, $v = i-3k$. Then $|u \times v|$ is: (1)
- A. 115 B. $\sqrt{115}$
C. 125 D. $\sqrt{125}$
-
43. Equation of tangent line to circle $x^2+y^2+2x-4y=0$ at point $(-2,4)$ is: (1)
- A. $2x - y + 8 = 0$ B. $x + 2y - 6 = 0$
C. $x - 2y - 10 = 0$ D. $x - 2y + 10 = 0$
-
44. If a translation T takes the origin to $(0,4)$, then the image of line $2x - y = 3$ under T is: (1)
- A. $2x - y = -1$ B. $2x - y = 1$
C. $2x + y = -1$ D. $2x + y = 1$
-
45. Image of circle $(x-1)^2+(y+2)^2=4$ when rotated by $3\pi/2$ about origin is: (1)
- A. $(x+2)^2+(y-1)^2=4$ B. $(x-2)^2+(y+1)^2=4$
C. $(x+1)^2+(y+2)^2=4$ D. $(x+2)^2+(y+1)^2=4$
-
46. Which of the following is a discrete data? (1)
- A. Volume of water in a glass B. Number of students in the class
C. Temperature in the classroom D. Time taken to complete a test
-
47. The following table presents scores of students. What is the middle quartile? (Table: Score 31-40:10, 41-50:9, 51-60:15, 61-70:12, 71-80:10, 81-90:4, 91-100:2) (1)
- A. 55.5 B. 50.5
C. 60.5 D. 58.5
-
48. What is the coefficient of x^3y^5 in the expansion of $(2x - y)^8$? (1)
- A. 448 B. 8
C. -448 D. -8
-
49. If two dice are rolled, probability of getting sum greater than 10 or doubles? (1)
- A. $4/9$ B. $2/9$
C. $1/4$ D. $1/9$

50. Which is the partial fraction form of $(x^2+3)/(x^3-x^2+x-1)$?

(1)

A. $2/(x-1) - (x+1)/(x^2+1)$

B. $2/(x-1) + (1-x)/(x^2+1)$

C. $2/(x-1) + (1-x)/(x+1)$

D. $2/(x-1) + (x+1)/(x^2+1)$

— END OF EXAMINATION —

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